Worksheet 7: Exoplanets and Habitable Worlds

Full solutions

Covers Lectures 13 and 14: transit and radial-velocity detection, atmospheric characterisation, the habitable zone, direct imaging, the Drake equation

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $k_B = 1.381 \times 10^{-23}$ J K^{-1}, $u = 1.661 \times 10^{-27}$ kg

1 AU = 1.496×10^{11} m, 1 pc = 3.086×10^{16} m, 1 yr = 3.156×10^{7} s, 1 d = 86,400 s, 1 rad = 206,265 arcsec

Sun: $M_\odot = 1.989 \times 10^{30}$ kg, $L_\odot = 3.828 \times 10^{26}$ W, $R_\odot = 6.957 \times 10^{8}$ m. Earth: $M_\oplus = 5.972 \times 10^{24}$ kg, $R_\oplus = 6.371 \times 10^{6}$ m, Bond albedo $A_B = 0.3$. Jupiter: $M_J = 1.898 \times 10^{27}$ kg

Hot Jupiter (a 51 Peg b analogue): $R_p = 8.6 \times 10^{7}$ m, $m_p \sin i = 0.5\, M_J$, $P = 4.23$ d, $a = 0.05$ AU, Sun-like host. Terminator: $T = 1500$ K, $\mu = 2.3$, $g = 8.6$ m s^{-2}, $n_H = 5$. K2-18 b: $m_p = 8.6\, M_\oplus$, $R_p = 2.6\, R_\oplus$

Rocky planet with a CO_2 atmosphere: $T = 300$ K, $\mu = 44$, $g = 9.81$ m s^{-2}. Spectrograph precision: ELODIE ~ 10 m s^{-1}; ESPRESSO 0.10 m s^{-1} (photon limit), ~ 0.5 m s^{-1} (long-term repeatability)

Kopparapu habitable-zone limits (Sun-like stars): runaway greenhouse $S_{\rm in,eff} = 1.06$, maximum CO_2 greenhouse $S_{\rm out,eff} = 0.35$. K dwarf: $L = 0.3\, L_\odot$, $M = 0.7\, M_\odot$. M dwarf (TRAPPIST-1-like): $L = 5 \times 10^{-4}\, L_\odot$. $t_{\rm MS,\odot} = 10$ Gyr

Reflected-light contrast at visible wavelengths: ~ 10^{-9} for a Jupiter analogue at 5 AU, ~ 10^{-10} for an Earth analogue; the Roman coronagraph reaches 10^{-8} to 10^{-9}. Data as in the Lecture 13 and 14 notes.

Problem 1 Transit depths and the geometric transit probability

The transit method measures the fraction of starlight a planet blocks, $\delta = (R_p/R_\star)^2$, but only for the small fraction of orbits aligned with our line of sight: a randomly oriented orbit transits with probability $p \approx R_\star/a$. These two numbers together explain what the transit catalogue contains, and what it misses.

(a) Compute the transit depth of the hot Jupiter in front of its Sun-like host.

Solution.

$$\delta = \left(\frac{R_p}{R_\star}\right)^2 = \left(\frac{8.6 \times 10^7}{6.957 \times 10^8}\right)^2 = (0.12362)^2 = 1.5281 \times 10^{-2}.$$

$\delta \approx 1.5\%$: a hot Jupiter dims a Sun-like star by more than one part in a hundred, which is why ground-based photometry could detect these transits from the start.

(b) Repeat for an Earth analogue transiting a Sun-like star, and express the result in parts per million.

Solution.

$$\delta = \left(\frac{6.371 \times 10^6}{6.957 \times 10^8}\right)^2 = (9.1577 \times 10^{-3})^2 = 8.3863 \times 10^{-5},$$

or about 84 ppm.

$\delta \approx 84$ ppm, about 180 times shallower than the hot-Jupiter transit. Detecting it needs space-based photometry at the 10 ppm level, which is the Kepler and PLATO regime.

(c) Compute the geometric transit probability for the Earth analogue at 1 AU and for the hot Jupiter at 0.05 AU, and the ratio of the two.

Solution.

$$p_\oplus = \frac{R_\star}{a} = \frac{6.957 \times 10^8}{1.496 \times 10^{11}} = 4.6504 \times 10^{-3},$$

about one in 215 randomly oriented orbits. For the hot Jupiter,

$$p_{\mathrm{HJ}} = \frac{6.957 \times 10^8}{0.05 \times 1.496 \times 10^{11}} = 9.3008 \times 10^{-2},$$

about one in eleven. The ratio is $p_{\mathrm{HJ}}/p_\oplus = (1\ \mathrm{AU})/(0.05\ \mathrm{AU}) = 20$.

$p_\oplus \approx 0.5\%$ against $p_{\mathrm{HJ}} \approx 9\%$: the close-in planet is 20 times more likely to transit, purely from geometry.

(d) The transit catalogue is dominated by short-period planets and by planets of M dwarfs. Using your results from parts (a) to (c), explain in two or three sentences why this does not mean that such planets are intrinsically the most common.

Solution. The transit probability $R_\star/a$ strongly favours close-in planets, and both the probability and the depth $(R_p/R_\star)^2$ grow when the star is small, so M-dwarf planets are easier to find in both respects. The catalogue therefore over-represents short-period and M-dwarf systems by construction. Occurrence rates only follow after correcting for these selection effects, and the corrected Kepler statistics show that small planets on wider orbits are far more common than hot Jupiters.

The catalogue reflects detection geometry, not intrinsic occurrence; the bias must be modelled out before population statements are made.

Problem 2 Radial-velocity masses and the density of K2-18 b

A planet and its star orbit their common centre of mass, so the star moves with a radial-velocity semi-amplitude

$$K_\star = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}}$$

for $m_p \ll M_\star$ and a circular orbit. Combining an RV mass with a transit radius gives the bulk density, the first handle on what a planet is made of.

(a) Compute $K_\star$ for the 51 Peg b analogue ($m_p \sin i = 0.5\,M_J$, $P = 4.23$ d) around a 1 $M_\odot$ star, and compare it with the ~ 10 m s^{-1} precision of ELODIE, the spectrograph that found 51 Peg b.

Solution. With $P = 4.23 \times 86{,}400 = 3.6547 \times 10^5$ s,

$$\left(\frac{2\pi G}{P}\right)^{1/3} = \left(\frac{4.1934 \times 10^{-10}}{3.6547 \times 10^5}\right)^{1/3} = (1.1474 \times 10^{-15})^{1/3} = 1.0469 \times 10^{-5},$$

and $M_\odot^{2/3} = (1.989 \times 10^{30})^{2/3} = 1.5816 \times 10^{20}$, so with $m_p \sin i = 9.490 \times 10^{26}$ kg,

$$K_\star = 1.0469 \times 10^{-5} \times \frac{9.490 \times 10^{26}}{1.5816 \times 10^{20}} = 62.82 \text{ m s}^{-1}.$$

$K_\star \approx 63$ m s^{-1}, six times the ELODIE precision: a close-in giant was the easiest possible first detection, and the short period meant the full orbit repeated every four days.

(b) Compute $K_\star$ for an Earth analogue ($m_p = M_\oplus$, $P = 1$ yr) around a 1 $M_\odot$ star, and compare it with the ESPRESSO photon-noise limit and long-term repeatability.

Solution. With $P = 3.156 \times 10^7$ s,

$$\left(\frac{2\pi G}{P}\right)^{1/3} = (1.3287 \times 10^{-17})^{1/3} = 2.3685 \times 10^{-6},$$

so

$$K_\star = 2.3685 \times 10^{-6} \times \frac{5.972 \times 10^{24}}{1.5816 \times 10^{20}} = 8.943 \times 10^{-2} \text{ m s}^{-1},$$

about 9 cm s^{-1}.

$K_\star \approx 9$ cm s^{-1} sits at the ESPRESSO photon limit (10 cm s^{-1}) but a factor of five below the ~ 0.5 m s^{-1} long-term repeatability, which is set by stellar activity, not photons. The true Earth analogue remains beyond current RV capability.

(c) K2-18 b has $m_p = 8.6\,M_\oplus$ from radial velocities and $R_p = 2.6\,R_\oplus$ from transits. Compute its bulk density and compare it with Earth's mean density of 5514 kg m^{-3}.

Solution. The mass is $m_p = 8.6 \times 5.972 \times 10^{24} = 5.1359 \times 10^{25}$ kg and the radius is $R_p = 2.6 \times 6.371 \times 10^6 = 1.6565 \times 10^7$ m, so

$$\rho = \frac{m_p}{\frac{4}{3}\pi R_p^3} = \frac{5.1359 \times 10^{25}}{\frac{4}{3}\pi (1.6565 \times 10^7)^3} = \frac{5.1359 \times 10^{25}}{1.9038 \times 10^{22}} = 2697.7 \text{ kg m}^{-3}.$$

$\rho \approx 2700$ kg m^{-3}, half of Earth's density: the planet cannot be bare rock and must

carry a substantial volatile component, either an H_2 envelope over a rocky core or a deep water layer. The bulk density alone cannot distinguish the two.

(d) Explain in two or three sentences why a radial-velocity detection alone gives only a minimum mass, and why the combination with a transit removes the ambiguity.

Solution. The spectrograph measures only the line-of-sight projection of the stellar reflex motion, so the observable is $m_p \sin i$ and the inclination i is unknown. A transiting planet must cross the stellar disk, which forces $i \approx 90^\circ$ and $\sin i \approx 1$, so the RV minimum mass becomes the true mass. The transit simultaneously supplies R_p, and mass plus radius give the bulk density.

RV alone measures $m_p \sin i$; a transit pins $\sin i \approx 1$ and adds the radius, turning a minimum mass into a density.

Problem 3 Atmospheric scale heights and transmission spectroscopy

During a transit, a small fraction of the starlight filters through the planet's terminator, and absorbing gases make the planet look larger at their wavelengths. The size of the effect is set by the atmospheric scale height $H = k_B T/(\mu\, u\, g)$, and the fractional change of the transit depth across a strong line is $\Delta\delta/\delta \approx 2n_H H/R_p$ for n_H scale heights of modulation.

(a) Compute the scale height of the hot Jupiter's terminator (T = 1500 K, μ = 2.3, g = 8.6 m s^{-2}).

Solution.

$$H = \frac{k_B T}{\mu\, u\, g} = \frac{(1.381 \times 10^{-23})(1500)}{(2.3)(1.661 \times 10^{-27})(8.6)} = \frac{2.0715 \times 10^{-20}}{3.2855 \times 10^{-26}} = 6.3050 \times 10^{5}\ \text{m},$$

about 630 km.

$H \approx 630$ km: hot, light, and weakly bound. A high temperature, an H_2/He mean molecular weight of 2.3, and the low gravity of an inflated giant puff the atmosphere up to a thickness that transmission spectroscopy can see.

(b) Compute the scale height of the rocky planet's CO_2 atmosphere (T = 300 K, μ = 44, g = 9.81 m s^{-2}), and the ratio of the two scale heights.

Solution.

$$H = \frac{(1.381 \times 10^{-23})(300)}{(44)(1.661 \times 10^{-27})(9.81)} = \frac{4.1430 \times 10^{-21}}{7.1695 \times 10^{-25}} = 5778.6\ \text{m},$$

about 5.8 km. The ratio is $6.3050 \times 10^5/5778.6 = 109.1$.

$H \approx 5.8$ km, a factor of ≈ 109 below the hot Jupiter: the colder temperature and the 19 times heavier molecule compress the atmosphere, and the transmission signal shrinks with it.

(c) For the hot Jupiter, compute the fractional modulation $\Delta\delta/\delta = 2n_H H/R_p$ with $n_H = 5$, then the absolute change in transit depth $\Delta\delta$ using the hot-Jupiter transit depth $\delta = 1.528 \times 10^{-2}$ of Problem 1(a), in ppm.

Solution.

$$\frac{\Delta\delta}{\delta} = \frac{2 \times 5 \times 6.3050 \times 10^5}{8.6 \times 10^7} = 7.3314 \times 10^{-2},$$

so

$$\Delta\delta = 7.3314 \times 10^{-2} \times 1.5281 \times 10^{-2} = 1.1203 \times 10^{-3},$$

about 1120 ppm.

$\Delta\delta \approx 1120$ ppm: a signal this large is why hot Jupiters became the first exoplanets with characterised atmospheres, within reach even of pre-JWST instruments. For the rocky planet the same estimate falls below 10 ppm and demands stacked multi-transit observations.

(d) JWST observed the rocky planet TRAPPIST-1 b in transmission and found a featureless spectrum, and its 15 μm secondary eclipse gave a dayside brightness temperature of $\approx$ 503 K, matching the 508 K bare-rock prediction. Explain in two or three sentences why the flat transmission spectrum alone could not settle whether the planet has an atmosphere, and what the eclipse measurement added.

Solution. A flat transmission spectrum is degenerate: an airless planet, a high-mean-molecular-weight atmosphere with a small scale height, and a light atmosphere hidden below high clouds all produce nearly featureless spectra at current precision. The secondary-eclipse brightness temperature breaks the degeneracy because a thick atmosphere would redistribute heat to the night side and lower the dayside temperature. The measured 503 K matches the zero-redistribution bare-rock value, so TRAPPIST-1 b has no thick atmosphere.

Transmission constrains the scale height, not the existence of an atmosphere; thermal emission tests heat redistribution and can identify a bare rock.

Problem 4 Equilibrium temperature and the habitable zone

A planet's equilibrium temperature follows from the balance of absorbed starlight and emitted thermal radiation,

$$T_{\rm eq} = \left[\frac{L_\star(1 - A_{\rm B})}{16\pi\sigma\epsilon d^2}\right]^{1/4},$$

and the habitable zone is the strip where the effective stellar flux $S_{\rm eff}$ lies between the runaway-greenhouse and maximum-CO_2-greenhouse limits, at orbital distances

$$d = (1\ \text{AU})\sqrt{\frac{L_\star/L_\odot}{S_{\rm eff}}}.$$

(a) Compute Earth's equilibrium temperature ($A_{\rm B} = 0.3$, $\epsilon = 1$, $d = 1$ AU), and reconcile it with the observed mean surface temperature of 288 K.

Solution.

$$T_{\rm eq} = \left[\frac{(3.828 \times 10^{26})(0.7)}{16\pi(5.670 \times 10^{-8})(1.496 \times 10^{11})^2}\right]^{1/4}$$

$$= \left[\frac{2.6796 \times 10^{26}}{6.3785 \times 10^{16}}\right]^{1/4} = (4.2010 \times 10^9)^{1/4} = 254.6\ \text{K}.$$

$T_{eq} \approx 255$ K, which is 33 K below the observed 288 K: the difference is the greenhouse effect of H_2O and CO_2. Equilibrium temperature is a radiation balance, not a surface climate.

(b) Compute the inner and outer habitable-zone distances for the Sun using $S_{in,eff} = 1.06$ and $S_{out,eff} = 0.35$, then the S_{eff} received by Venus (0.72 AU) and Mars (1.52 AU), and place both planets relative to the zone.

Solution.

$$d_{in} = \frac{1\ \text{AU}}{\sqrt{1.06}} = 0.9713\ \text{AU}, \qquad d_{out} = \frac{1\ \text{AU}}{\sqrt{0.35}} = 1.6903\ \text{AU}.$$

The received flux scales as $S_{eff} = (d/1\ \text{AU})^{-2}$: Venus receives $1/0.72^2 = 1.929$, well above the runaway limit of 1.06; Mars receives $1/1.52^2 = 0.4328$, above the outer limit of 0.35.

The Sun's habitable zone spans 0.97 to 1.69 AU. Venus sits far inside the inner edge, consistent with its runaway state; Mars sits inside the outer edge, so its frozen state reflects its small mass and lost atmosphere rather than its orbit.

(c) Compute the habitable-zone boundaries for the K dwarf ($L = 0.3\,L_\odot$) and the M dwarf ($L = 5 \times 10^{-4}\,L_\odot$), then the main-sequence lifetime of the K dwarf ($M = 0.7\,M_\odot$) from $t_{MS} \approx t_{MS,\odot}(M_\star/M_\odot)^{-2.5}$.

Solution. For the K dwarf,

$$d_{in} = \sqrt{\frac{0.3}{1.06}} = 0.5320\ \text{AU}, \qquad d_{out} = \sqrt{\frac{0.3}{0.35}} = 0.9258\ \text{AU};$$

for the M dwarf,

$$d_{in} = \sqrt{\frac{5 \times 10^{-4}}{1.06}} = 0.02172\ \text{AU}, \qquad d_{out} = \sqrt{\frac{5 \times 10^{-4}}{0.35}} = 0.03780\ \text{AU},$$

inside the orbit of Mercury. The K-dwarf lifetime is

$$t_{MS} = 10 \times 0.7^{-2.5} = 10 \times 2.4392 = 24.39\ \text{Gyr}.$$

The habitable zone shrinks and moves inward as $\sqrt{L_\star}$, from ~ 1 AU scales at the Sun to a few hundredths of an AU at TRAPPIST-1, while the stellar lifetime grows steeply: the 0.7 $M_\odot$ K dwarf outlives the Sun by a factor of 2.4.

(d) M dwarfs offer the deepest transits, the largest transit probabilities in the habitable zone, and the longest lifetimes, yet they are not automatically the best hosts for life. Name the main physical drawbacks of an M-dwarf habitable zone in two or three sentences.

Solution. At a few hundredths of an AU every habitable-zone planet is tidally locked, with a permanent day and night side. During the star's long pre-main-sequence phase the star was far more luminous, so the planets that now sit in the habitable zone spent their first ~ 10^8 yr inside the runaway-greenhouse boundary and may have lost their water before the star settled down. M dwarfs also stay magnetically active for Gyr, and their flares and winds erode atmospheres at such close orbital distances.

Tidal locking, the luminous pre-main-sequence phase, and sustained flare activity all work against M-dwarf habitability; K dwarfs avoid most of these problems and may be the most favourable hosts.

Problem 5 Direct imaging and the Drake equation

Direct imaging separates the planet's photons from the star's on the detector. By the small-angle formula the separation on the sky is $\theta = a/d$, which by the definition of the parsec is $\theta[\text{arcsec}] = a[\text{AU}]/d[\text{pc}]$. The final problem turns from detection to the question the whole field serves: how many inhabited worlds are out there?

(a) Compute the angular separation of an Earth analogue ($a = 1$ AU) and a Jupiter analogue ($a = 5$ AU) around a Sun-like star at $d = 10$ pc, in arcseconds.

Solution.

$$\theta_\oplus = \frac{1.496 \times 10^{11}}{10 \times 3.086 \times 10^{16}} = 4.8477 \times 10^{-7}\ \text{rad} = 4.8477 \times 10^{-7} \times 206{,}265 = 0.09999\ \text{arcsec},$$

and the Jupiter analogue at 5 AU sits at $5 \times 0.09999 = 0.5000$ arcsec.

0.1 and 0.5 arcsec are resolvable separations for large telescopes; the difficulty of direct imaging is not the angle but the contrast, 10^{-10} for the Earth analogue against 10^{-9} for the Jupiter analogue, which is why the Roman coronagraph (10^{-8} to 10^{-9}) targets giants while Earth analogues wait for the next generation.

(b) Using the habitable-zone boundaries from Problem 4 (0.9713 to 1.690 AU for the Sun, 0.02172 to 0.03780 AU for the M dwarf), compute the angular extent of the Sun's habitable zone and of the M dwarf's habitable zone, both at 10 pc.

Solution. For the Sun-like star, $\theta = a[\text{AU}]/d[\text{pc}]$ gives $0.9713/10 = 0.09713$ arcsec at the inner edge and $1.6903/10 = 0.16903$ arcsec at the outer edge. For the M dwarf the same formula gives

$$\theta_{\text{in}} = \frac{0.02172}{10} = 2.172 \times 10^{-3}\ \text{arcsec}, \qquad \theta_{\text{out}} = \frac{0.03780}{10} = 3.780 \times 10^{-3}\ \text{arcsec},$$

a few milliarcseconds.

The Sun-like habitable zone spans a tenth of an arcsecond, within reach of a coronagraphic space telescope; the M-dwarf habitable zone collapses to milliarcseconds, out of reach for direct imaging. The two detection methods divide the sky: transits own the M dwarfs, direct imaging owns the Sun-like stars.

(c) The Drake equation factorises the number of communicating civilisations as

$$N = R_\star f_p n_e f_l f_i f_c L,$$

and only the first three factors are measured. Model the four unconstrained factors as independent and log-uniform, each with $\log_{10}$ distributed uniformly on $[-10, 0]$, and take the measured prefactor as $\log_{10}(R_\star f_p n_e) \approx 0$. Compute the mean and standard deviation of $\log_{10} N$, and the number of standard deviations separating the mean from $N = 1$.

Solution. Each factor contributes a mean of −5 and a variance of $10^2/12 = 8.3333$, so

$$\langle \log_{10} N \rangle = 4 \times (-5) = -20, \qquad \sigma = \sqrt{4 \times 8.3333} = \sqrt{33.333} = 5.7735.$$

The value $N = 1$ corresponds to $\log_{10} N = 0$, which lies

$$\frac{0 - (-20)}{5.7735} = 3.464$$

standard deviations above the mean.

Under honestly broad priors the expectation sits twenty decades below one civilisation, and $N = 1$ is a 3.5σ upward excursion: "we are alone in the galaxy" is fully consistent with current knowledge. The point is not the specific number but that the output distribution restates the priors, which is why the Drake equation cannot function as an estimator.

(d) The lectures presented the Drake equation as a framing tool, not an estimator. Give the main reasons in two or three sentences.

Solution. The factors are not independent: biospheres co-evolve with their atmospheres and planets, so factorising the history into separate probabilities discards the connections between them. Four of the seven factors are unconstrained by many orders of magnitude, so any product restates the chosen priors rather than a measurement. And the single known origin of life is observed from the inside, so anthropic selection biases every extrapolation from the Earth case.

Coupled factors, order-of-magnitude ignorance, and anthropic bias make the equation a map of what we do not know, which is precisely its pedagogical value.